

Effects of personalized nutrition on cardiometabolic biomarkers in adults with overweight or obesity: a systematic review and meta-analysis of randomized controlled trials

Received: 17 March 2026

Accepted: 28 April 2026

Published online: 09 May 2026

Cite this article as: Duc T.Q., Khanh N., Khoa D.D. *et al.* Effects of personalized nutrition on cardiometabolic biomarkers in adults with overweight or obesity: a systematic review and meta-analysis of randomized controlled trials. *Nutr Metab (Lond)* (2026). <https://doi.org/10.1186/s12986-026-01130-3>

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Title: Effects of Personalized Nutrition on Cardiometabolic Biomarkers in Adults with Overweight or Obesity: A Systematic Review and Meta-analysis of Randomized Controlled Trials

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Abstract

Purpose

Personalized nutrition (PN) shows promise for improving cardiometabolic health; however, evidence on its summarized methodologies and effectiveness remains limited and inconsistent. This review evaluates the effects of PN interventions on cardiometabolic outcomes in adults with overweight or obesity.

Methods

We searched five databases for randomized controlled trials (RCTs) published up to March 2026. Eligible RCTs included at least one biological component to guide personalized dietary advice. Data were pooled using a random-effects meta-analysis, with heterogeneity assessed by Cochran's Q statistic and quantified using I^2 . Risk of bias and certainty of evidence were assessed.

Results

Fifteen RCTs met the inclusion criteria, each comparing PN with standard, non-personalized dietary interventions. PN implementation varied substantially in the quantity and type of individual-level features, algorithms applied, and personalization strategies used. No significant differences were observed between PN and controls in energy intake or macronutrient composition. 46.7% employed specific dietary patterns as controls or withheld nutritional counseling. Pooled analyses showed slightly greater reductions in body weight and body fat in the PN group compared with controls, but no effects on body mass index, waist circumference, lipid profile, or glycemic markers. GRADE certainty ranged from moderate to very low across the assessed outcomes.

Conclusion

PN interventions modestly reduce body weight and body fat. Importantly, the absence of a standardized PN framework hampers evaluation and comparison of the effectiveness of nutritional personalization strategies. Future research should aim to establish standardized PN definitions, harmonized individual-level features, and transparent reporting.

Registry and registry number for systematic reviews or meta-analyses:

CRD420251134855

Keywords: *Personalized Nutrition, Systematic Review, Meta-Analysis, Cardiometabolic Health, Obesity, Nutrigenomics*

Abbreviations

AI	Artificial Intelligence
BMI	Body Mass Index
CI	Confidence Interval
CVD	Cardiovascular disease
CGM	Continuous Glucose Monitoring
FTO	Fat Mass And Obesity-Associated
GRADE	Grading of Recommendations, Assessment, Development, and Evaluations
HbA1c	Hemoglobin A1c
HDL-C	High-density lipoprotein cholesterol
LDL-C	Low-density lipoprotein cholesterol
MD	Mean Difference
ML	Machine Learning
PICOS	Participants, Interventions, Control, Outcomes, Study Design
PN	Personalized Nutrition
PRISMA	Preferred Reporting Items For Systematic Reviews And Meta-Analyses
RCTs	Randomized Controlled Trials
ROS	Reactive Oxygen Species
SD	Standard Deviation

Introduction

Cardiovascular disease (CVD) was the leading cause of global mortality in 2021 [1], and 36.7 % of CVD mortality was attributable to dietary risk factors [2]. Traditionally, dietary guidelines have been conveyed to the general population through simple visual aids designed to address broad health outcomes [3]. Although population-level dietary guidelines provide an essential foundation for public health nutrition, they may not fully account for interindividual variability in biological, behavioral, and environmental factors that can influence dietary responses [4, 5]. This variability, compounded by broader sociocultural influences, poses significant challenges to advancing diet-based strategies for CVD prevention and care. In response, the concept of personalized nutrition (PN) has emerged, extending the principles of precision medicine through advanced biological and technological tools to tailor dietary recommendations to individual-level features [5].

A universally accepted definition of PN remains elusive, with the concept being described in various ways across the literature. Terms such as "precision nutrition," "individualized nutrition," and "nutritional genomics" are often used interchangeably, reflecting overlapping, yet distinct, frameworks [6, 7]. Despite these terminological differences, the primary objective of all these approaches is consistent: to address individual variability by integrating diverse, multi-dimensional data in order to characterize an individual's health status, assess disease risk, and deliver personalized dietary recommendations aimed at optimizing health outcomes. Over the past two decades, research in PN has increasingly focused on integrating lifestyle factors with biomedical parameters, chiefly genetic and blood biomarkers, to personalize dietary advice [4]. More recently, the inclusion of microbiome profiling has expanded the scope of genotype-based analyses [4, 8]. Parallel to these developments, advances in biomarker discovery, the proliferation of wearable technologies, and innovations in noninvasive monitoring devices have dramatically increased the volume and real-time nature of data available for personalizing dietary advice. Hitherto, the application of artificial intelligence (AI) and machine learning (ML) has advanced PN by integrating complex health

data to predict individual metabolic responses and to generate and deliver highly specific, data-driven dietary and lifestyle recommendations [9].

Although PN shows potential for improving cardiometabolic outcomes through personalized approaches [9, 10], the current body of evidence remains limited, and existing findings are often inconsistent or inconclusive. For instance, the ZOE 2022 algorithm-based PN, which integrates food characteristics, postprandial glucose and triglyceride responses to foods, microbiome composition, cardiovascular disease risk, and health history, has demonstrated improvements in body weight, waist circumference, HbA1c, and microbiome diversity [11]. Conversely, the PREVENTOMICS study, which employed a PN framework based on five core physiological processes—lipid metabolism, carbohydrate metabolism, systemic inflammation, oxidative stress, and microbiota status—did not show significant reductions in fat mass or improvements in lipid profiles or insulin sensitivity compared to control arm [12]. Furthermore, a recent review found that PN interventions with dietary counseling improved CVD risk factors in at-risk individuals [10]. However, the review did not outline minimum requirements for which individual-level features constitute PN in the included trials, leaving the quality and strength of evidence uncertain. In practice, interventions labeled as PN can range from basic tailoring based on self-reported dietary habits to advanced algorithms that integrate genetic, metabolic, or microbiome data. Establishing minimum requirements helps distinguish PN from general dietary advice and enables more reliable comparisons of its cardiometabolic effects across trials. In addition, no meta-analysis of PN in overweight or obese populations has yet been conducted, restricting the ability to translate current findings into actionable dietary guidelines and policy recommendations. To address these gaps, by establishing minimum requirements for PN, our systematic review and meta-analysis evaluate whether PN interventions lead to greater improvements in anthropometric measures, lipid profiles, and glycemic indices compared with control diets, synthesizing evidence from randomized controlled trials (RCTs) in overweight and obese adults.

Materials And Methods

This systematic review was conducted following the Cochrane Handbook for Systematic Reviews of Interventions and is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. The study protocol was registered with PROSPERO (CRD420251134855).

Eligibility criteria

The eligibility criteria were developed based on the Population, Intervention, Comparator, and Outcome (PICO) framework, as detailed in **Supplementary S1**. Eligible RCTs were defined as those implementing a personalized dietary intervention tailored to participant characteristics, with a comparison group receiving standard care, no intervention, or a non-personalized dietary consultation. All comparator groups are herein referred to as the “control group”. Participants had to be free-living adults (≥ 18 years) living with overweight or obesity, as defined by the criteria used in each individual study.

Given the absence of a universally accepted definition of PN, we adopted the following operational definition: PN aims to maintain or improve health by providing tailored dietary and lifestyle recommendations based on genetic, phenotypic, medical history, dietary intake, lifestyle behaviors, nutritional status, and the composition of the gut microbiome [4, 8, 13]. Because there is currently no consensus on which and how many individual-level features are required, we specified that eligible RCTs must include at least one biological component—such as a genetic marker (e.g., fat mass and obesity-associated [FTO] gene), a metabolic biomarker (e.g., blood glucose, cholesterol), or a gut microbiome profile—to guide the development of personalized dietary advice, whether generated by ML algorithms, healthcare professionals, or both. This criterion ensures a minimum level of biological personalization, distinguishing such interventions from standard dietary recommendations based solely on lifestyle or behavioral factors. Additionally, RCTs may incorporate these biological components alongside other individual-level characteristics, such as anthropometric parameters (e.g., height, weight) and lifestyle factors (e.g., dietary intake, medical history, physical activity).

Search strategy

A systematic search was conducted across four electronic databases—PubMed, Scopus, Web of Science, and ClinicalTrials.gov—from their inception to March 01, 2026, to identify RCTs evaluating the effects of PN on anthropometric outcomes, glycemic control, and lipid profiles. In addition, preprint servers such as MedRxiv were screened to capture unpublished or ongoing studies, and the reference lists of included articles and relevant systematic reviews from the initial search were manually examined. No language restrictions were applied. The complete search strategy, including all terms and search strings, is detailed in **Supplementary S2**.

Data extraction and collection

References were imported into a reference management software (Zotero Desktop) for eligibility assessment. Following both automatic and manual removal of duplicates, two

investigators (NTHH and NDK) independently screened titles and abstracts, followed by full-text articles, based on predefined inclusion and exclusion criteria. Discrepancies were resolved by consultation with a third reviewer (TQD).

Data from eligible studies were extracted using a standardized spreadsheet template (Microsoft Excel, Microsoft Corp.) and included the following variables: first author, year of publication, participant age, total sample size, sex distribution, study country and population, study duration, conflicts of interest, funding source, individual-level features (physical activity, microbiome profiling, metabolomic profiling, medical history, genetic markers, dietary intake, anthropometric parameters), use of ML algorithms, adherence evaluation, use of digital applications to support the PN intervention, energy intake (kcal), macronutrient composition (carbohydrate, protein, fat), and summary statistics for outcome measures at baseline and at the end of the intervention. Where available, change-from-baseline values and corresponding standard deviations (SDs) for both the PN and control groups were used to calculate the overall effect size as mean difference (MD). When outcomes were reported as medians with interquartile ranges or means with 95% confidence intervals (CIs), the respective means and SDs were estimated using appropriate statistical methods [14, 15]. If essential data were missing and could not be calculated from the reported information, study authors were contacted to obtain the required data. Studies were excluded if no data could be retrieved. For trials with multiple publications, data were extracted from the report with the longest follow-up duration or the available outcomes most relevant to this review. All extracted data were independently cross-checked by two investigators, and any discrepancies were resolved through consensus.

Risk of bias assessment

Risk of bias (RoB) in eligible studies was independently assessed by two investigators (DDK and NVK) using the Cochrane RoB 2.0 tool [16]. Discrepancies between investigators were resolved through discussion with a third reviewer (TQD). RoB assessments were summarized and presented using the robvis data visualization tool [17].

Data synthesis

Study and intervention characteristics of all included trials were summarized in tables and figures. Significant differences in age, sample size, sex distribution, study duration, and changes in caloric and macronutrient intake were assessed using a two-tailed t-test. Prior to meta-analysis, all outcome data were standardized to uniform units, with the exception of BMI (kg/m^2), body weight (kg), body fat (%), and insulin ($\mu\text{IU/mL}$), which did not require conversion. Glucose values were converted from mmol/L to mg/dL using the factor 18.02

(mg/dL = mmol/L $\times$ 18.02). Triglyceride values were converted using the factor 88.57 (mg/dL = mmol/L $\times$ 88.57). Total cholesterol, LDL-C, and HDL-C values were converted using the factor 38.67 (mg/dL = mmol/L $\times$ 38.67) [18].

A meta-analysis was performed when quantitative data for an outcome were available from at least two studies with sufficiently comparable measures to allow pooling. Given the anticipated heterogeneity in populations (e.g., ethnic background, variation in the combinations of individual-level features, and the modes of delivery of personalized diets), effect sizes were pooled using the inverse-variance method with a random-effects model. To account for both within- and between-study variance, the Sidik–Jonkman estimator with Hartung–Knapp adjustment was applied. MDs in outcomes from baseline, along with their 95% confidence intervals (CIs), were pooled. Statistical heterogeneity was assessed using Cochran’s Q (χ^2) test and quantified using the I^2 statistic; the p-value for heterogeneity shown in the forest plots was derived from Cochran’s Q test, with $p < 0.10$ considered statistically significant in accordance with Cochrane Handbook guidance [19]. I^2 was interpreted as follows: 0–40% as minimal, 30–60% as moderate, 50–90% as substantial, and 75–100% as considerable heterogeneity [19].

The performance of the meta-analysis may be limited when only a small number of trials are included. Consequently, following the rule of thumb used in small-study effect testing, sensitivity analyses were conducted by sequentially removing individual studies and recalculating effect sizes, but only when at least ten studies were available for a given outcome [20]. Similarly, potential sources of heterogeneity were examined through prespecified subgroup analyses, conducted when a minimum of ten studies contributed to the outcome [21]. Differences between subgroups were assessed using tests for subgroup interaction, and residual heterogeneity (I^2) was examined within subgroups. The subgroup factors included key features of PN interventions: physical activity (yes/no), gut microbiome profiling (yes/no), metabolomics profiling (yes/no), medical history (yes/no), genetic markers (yes/no), dietary intake (yes/no), and anthropometric parameters (yes/no). Additional subgroup analyses were conducted based on the use of machine learning algorithms to guide PN development (yes/no), adherence evaluation (yes/no), digital applications (yes/no), dietary control (specific vs. non-specific dietary patterns), economic status (developed vs. developing countries), study duration (≥ 20 vs. < 20 weeks), and study region (Asia vs. non-Asia).

In accordance with the Cochrane Handbook, visual inspection of funnel plots and Egger’s test were used to assess reporting bias when at least 10 studies were included for a given outcome

[22]. When publication bias was suspected (i.e., funnel plot asymmetry or a significant Egger's test, $P < 0.05$), the Duval and Tweedie trim-and-fill method was applied.

Statistical significance was set at two-sided $P < 0.05$, unless otherwise stated. Analyses and visualization were conducted in RStudio (version 2025.05.0 Build 496; RStudio, Boston, MA, USA) using the *ggplot2*, *meta*, and *metafor* packages.

Certainty of the evidence

The Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) framework was applied independently by two investigators (DDK and TTT) to assess the certainty of evidence from the meta-analyses [23]. Disagreements were resolved by consensus and, if necessary, through consultation with a third reviewer (TQD). The certainty of evidence for each outcome was classified into one of four levels: high, moderate, low, or very low [23]. The magnitude of effect was further interpreted for each outcome by considering the size and direction of the pooled estimate alongside the certainty of evidence and a minimally important difference, and was expressed using GRADE-consistent descriptive statements [24].

Results

Study selection

Supplementary S3 illustrates the PRISMA flow diagram. The search yielded 543 unique records after removal of duplicates. Following title and abstract screening and supplementary hand searching, 59 full-text articles were assessed for eligibility. Ultimately, 15 studies met the inclusion criteria and were included in this review [11, 12, 25-37]. Detailed reasons for full-text exclusions are provided in **Supplementary S4**.

Study characteristics

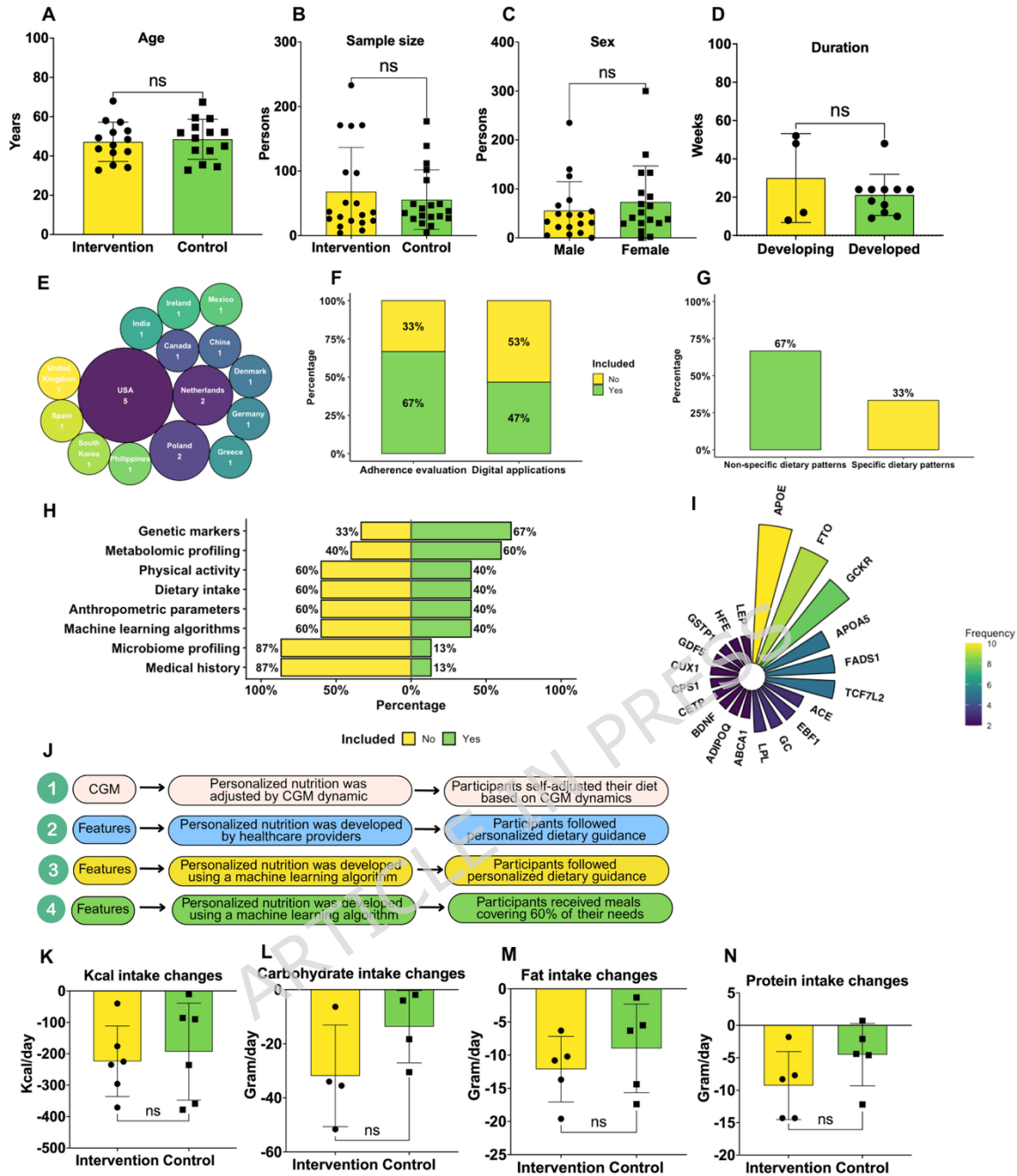


Figure 1. Characteristics of the included randomized controlled trials and features of personalized nutrition (PN) interventions.

(A) Mean age of participants in the intervention and control groups.

(B) Sample sizes of the intervention and control groups.

(C) Sex distribution across included studies.

(D) Study duration (weeks) in studies conducted in developing versus developed countries.

(E) Geographic distribution of the included studies.

(F) Proportion of included studies that assessed adherence and used digital applications.

(G) Types of control conditions used in the included studies.

(H) Individual-level features used to guide PN interventions.

(I) Frequency of genes targeted in genetically informed PN approaches.

(J) Summary of the main implementation steps of PN interventions in the included studies, including the source of personalization, the method used to develop PN, and how participants followed or received the intervention.

(K–N) Changes in total energy, carbohydrate, fat, and protein intake from baseline to the end of the intervention.

For panels A–D and K–N, bars represent mean values and dots represent individual studies; between-group comparisons were performed using two-sample t-tests, and ns indicates no statistically significant difference.

Abbreviations: PN, personalized nutrition; CGM, continuous glucose monitoring; ns, not significant.

The characteristics of the included studies are summarized in **Figure 1A–E** and **Supplementary S5**. A total of 15 RCTs, published between 2016 and 2025, were included, comprising 2,351 participants. The mean age was 47.2 ± 9.7 years in the PN groups and 48.5 ± 9.4 years in the control groups. PN group sizes ranged from 4 to 233 participants, while control group sizes ranged from 5 to 177. In terms of sex distribution, 978 participants were male and 1,281 were female; one study did not report sex-specific data [36]. Trial duration varied by geographic context: studies conducted in developing countries had a longer average duration (approximately 30 weeks) compared to those in developed countries (approximately 21 weeks). Trials were geographically diverse, with most conducted in the United States ($n = 5$), and others carried out in Europe, Asia, and North America, including India, Canada, South Korea, the Netherlands, the Philippines, Romania, Mexico, China, and Poland. Overall, no statistically significant differences were observed across studies in age, sample size, sex distribution, or study duration.

We also identified and summarized several methodological features of PN delivery, as presented in **Figure 1F–J** and **Supplementary S5**. First, 67% of studies assessed participant adherence, and 47% incorporated digital applications to support intervention delivery. Second, 33% of trials implemented a defined dietary pattern—such as a low-fat diet or calorie restriction—as the control condition [25, 27, 30, 33, 36], while 13.3% did not offer any dietary counseling to control participants [28, 29]. Third, 40% of studies employed machine learning algorithms to generate personalized dietary recommendations, using varying combinations of individual-level features. Genetic markers were the most commonly used (67%), whereas gut microbiome profiles and medical history were least frequently incorporated, each reported in only 13% of trials. Fourth, the most commonly targeted genes were APOE, FTO, and GCKR (see Supplementary S6 for details). Finally, four primary approaches to PN implementation were identified, classified by the source of

personalization—whether algorithm-based, practitioner-guided, or other. Specifically, Approach 1 was reported in one study [30], Approach 2 in eight studies [27-29, 31, 32, 34-36], Approach 3 in five studies [11, 25, 26, 33, 37], and Approach 4 in one study [12].

Finally, we summarized the changes in total energy and macronutrient intake between the PN and control groups in **Figure 1K–N** and **Supplementary S7**. Of the 15 included studies, six reported data on these measurements, and none observed significant differences between groups at the end of the intervention.

Risk of bias in included trials

Of the 15 included trials, six (40%) were judged to have an overall low risk of bias, while nine (60%) were rated as having some concerns. The risk of bias assessment across the five domains for each included study is presented in **Supplementary S8A**, while the overall summary of RoB judgments is shown in **Supplementary S8B**.

Anthropometrics outcomes

Figure 2 presents the effects of PN on anthropometric outcomes. Eighteen effect sizes were pooled for body weight, thirteen for BMI, twelve for body fat, and sixteen for waist circumference. The meta-analysis showed that PN interventions significantly reduced body weight (MD: -1.27 kg; 95% CI: -2.38 to -0.16 ; $p = 0.03$; $I^2 = 94.3\%$) and body fat percentage (MD: -0.44% ; 95% CI: -0.76 to -0.12 ; $p = 0.01$; $I^2 = 78.9\%$). In contrast, reductions in BMI (MD: -0.35 kg/m²; 95% CI: -0.92 to 0.22 ; $p = 0.21$; $I^2 = 90.5\%$) and waist circumference (MD: -1.09 cm; 95% CI: -2.73 to 0.55 ; $p = 0.18$; $I^2 = 90.8\%$) were not statistically significant.

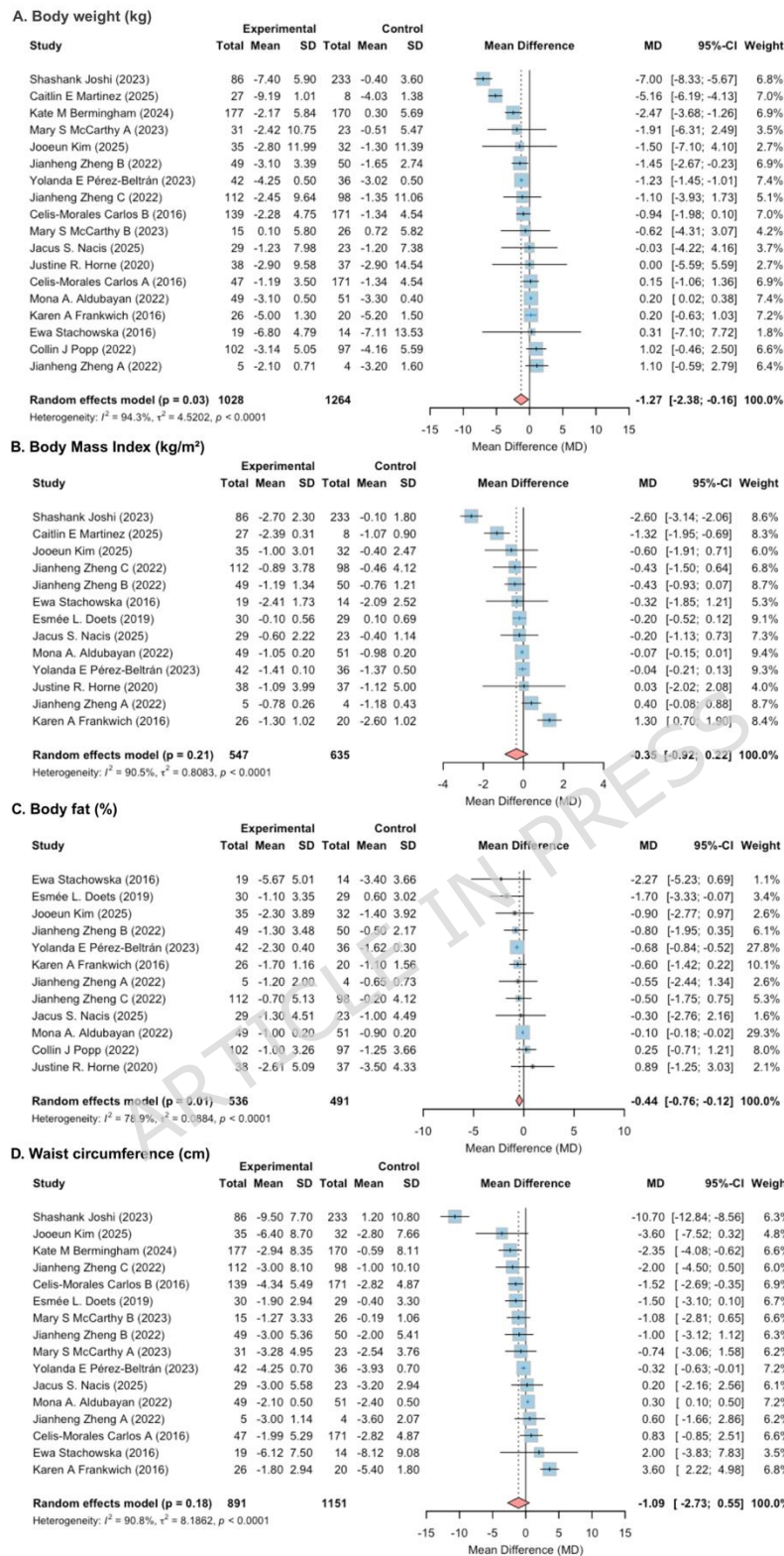
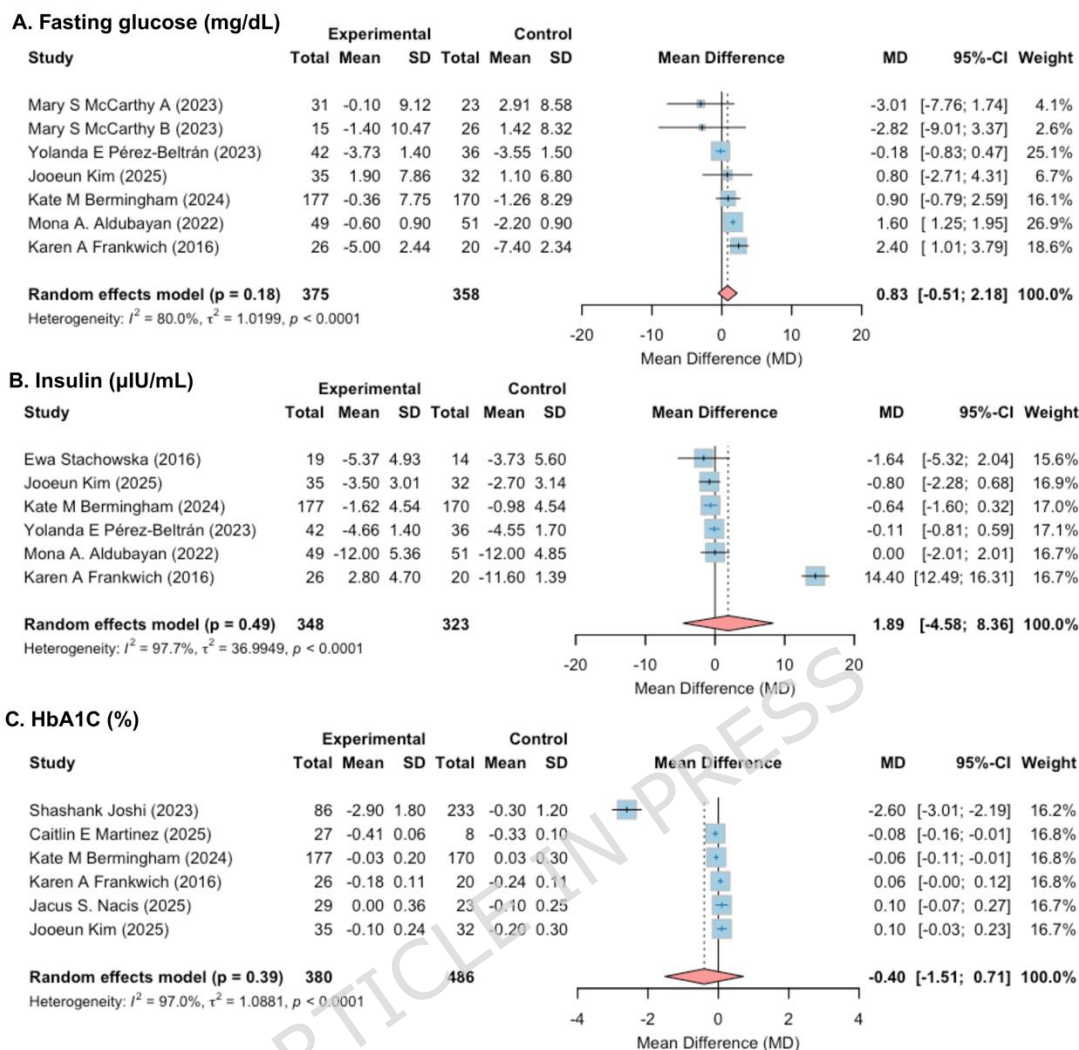


Figure 2. Forest plots displaying the effect of personalized nutrition in adults on (A) body weight (kg), (B) body mass index (kg/m²), (C) body fat (%), and (D) waist circumference (cm). Abbreviations: CI, confidence interval; MD, mean difference; SD, standard deviation; Total, total number of participants who completed the study.



Glycemic outcomes

Figure 3. Forest plots displaying the effect of personalized diet in adults on (A) fasting glucose (mg/dL), (B) Insulin ($\mu\text{IU/mL}$), and (C) HbA1c (%). Abbreviations: CI, confidence interval; MD, mean difference; SD, standard deviation; Total, total number of participants who completed the study.

Figure 3 presents the effects of PN on glycemic outcomes. Seven effect sizes were pooled for fasting glucose, six for insulin, and six for HbA1c. Overall, PN interventions showed no significant effects on fasting glucose (MD: 0.83 mg/dL; 95% CI: -0.51 to 2.18; $p = 0.18$; $I^2 = 80.0\%$), insulin levels (MD: 1.89 $\mu\text{IU/mL}$; 95% CI: -4.58 to 8.36; $p = 0.49$; $I^2 = 97.7\%$), or HbA1c (MD: -0.40%; 95% CI: -1.51 to 0.71; $p = 0.39$; $I^2 = 97.0\%$).

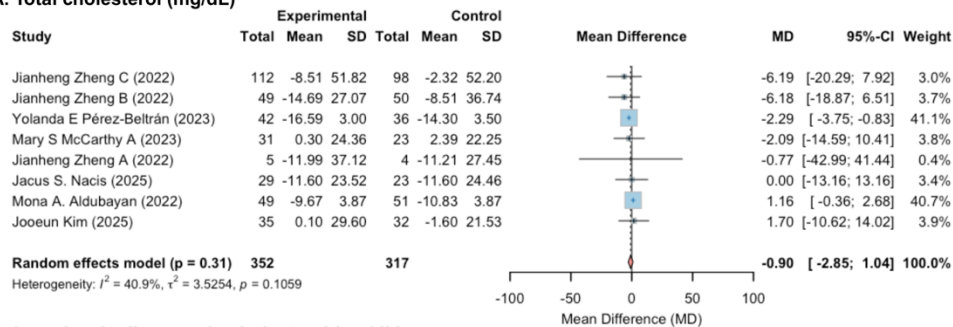
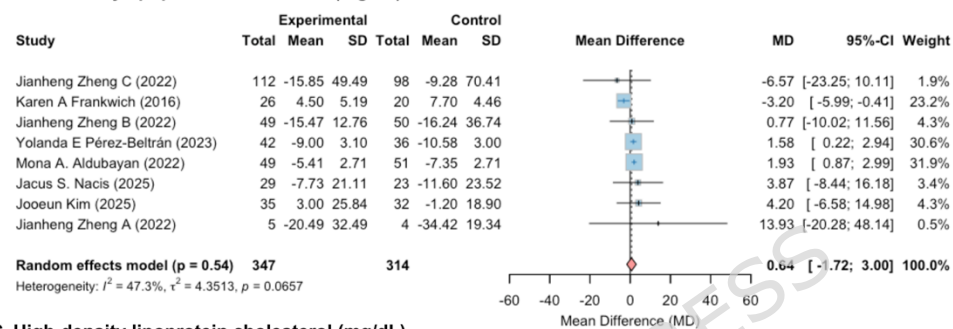
A. Total cholesterol (mg/dL)**B. Low-density lipoprotein cholesterol (mg/dL)****C. High-density lipoprotein cholesterol (mg/dL)**

Figure 4. Forest plots displaying the effect of personalized diet in adults on (A) total cholesterol (mg/dL), (B) low-density lipoprotein cholesterol (LDL-C, mg/dL), (C) high-

density lipoprotein cholesterol (HDL-C, mg/dL), and (D) triglycerides (mg/dL). Abbreviations: CI, confidence interval; MD, mean difference; SD, standard deviation; Total, total number of participants who completed the study.

Figure 4 presents the effects of PN on lipid profile outcomes. Eight effect sizes were pooled for total cholesterol and LDL-C, and nine for HDL-C and triglycerides. Overall, PN interventions did not produce statistically significant effects on total cholesterol (MD: -0.90 mg/dL; 95% CI: -2.85 to 1.04 ; $p = 0.31$; $I^2 = 40.9\%$), LDL-C (MD: 0.64 mg/dL; 95% CI: -1.72 to 3.00 ; $p = 0.54$; $I^2 = 47.3\%$), HDL-C (MD: -1.73 mg/dL; 95% CI: -4.94 to 1.49 ; $p = 0.25$; $I^2 = 95.4\%$), or triglycerides (MD: -2.22 mg/dL; 95% CI: -19.58 to 15.14 ; $p = 0.78$; $I^2 = 96.8\%$).

Subgroup analyses

The results of the subgroup analyses are presented in **Supplementary S9**. For body weight, no significant subgroup differences were identified, although significant reductions were observed in studies that did not incorporate anthropometric parameters, medical history, or microbiome profiling as guiding features. For BMI, no subgroup category showed a significant pooled reduction, although between-subgroup differences were observed for digital applications, metabolomic profiling, and genetic markers. For body fat, significant subgroup differences were observed for economic status, digital applications, and metabolomic profiling. Significant reductions in body fat were observed in studies conducted in Asia or developing countries and in interventions lasting fewer than 20 weeks. Similar reductions were also seen in trials that did not use digital applications, dietary intake, medical history, microbiome profiling, machine learning algorithms, as well as in studies incorporating genetic markers, and trials using non-specific dietary patterns as controls. For waist circumference, significant reductions were observed in studies incorporating medical history or microbiome information and in interventions lasting fewer than 20 weeks, whereas only genetic markers showed evidence of a significant between-subgroup difference.

Sensitivity analysis and Publication bias

Sensitivity analysis revealed no significant changes in heterogeneity, statistical significance, or the direction of the pooled effects for BMI and waist circumference (**Supplementary S10**). However, the effect on body weight was no longer statistically significant after omitting the study by Caitlin E. Martinez [30] (MD: -0.98 kg; 95% CI: -2.03 to 0.07 ; $I^2 = 92.6\%$).

Similarly, the reduction in body fat was rendered non-significant when the study by Yolanda E. Pérez-Beltrán [33] was excluded (MD: -0.31% ; 95% CI: -0.67 to 0.04 ; $I^2 = 10.8\%$). Visual inspection of funnel plots (**Supplementary S11**) showed no evidence of asymmetry, and Egger's test likewise indicated no publication bias for body weight ($p = 0.25$), BMI ($p = 0.25$), body fat ($p = 0.29$), or waist circumference ($p = 0.15$). Given the limited number of studies—fewer than ten—sensitivity analysis and publication bias assessment were not conducted for glycemic and lipid profile outcomes.

GRADE assessment

According to the GRADE framework, the certainty of evidence was rated as moderate for body fat, total cholesterol, and LDL-C; low for body weight, waist circumference, and fasting glucose; and very low for BMI, insulin, HbA1c, HDL-C, and triglycerides. In terms of magnitude of effect, PN likely results in a slight reduction in body fat percentage and likely results in little to no difference in total cholesterol and LDL-C. PN may result in a slight reduction in body weight and may result in little to no difference in waist circumference or fasting glucose. For BMI, insulin, HbA1c, HDL-C, and triglycerides, the evidence is very uncertain. These ratings were downgraded mainly because of inconsistency across studies, particularly for outcomes with substantial to very high heterogeneity, and for imprecision in outcomes with wide confidence intervals. The summary of findings for the GRADE assessment is available in the **Supplementary S12**.

Discussion

Principal findings

To our knowledge, this is the first meta-analysis to evaluate the effects of PN on cardiometabolic biomarkers in adults with overweight or obesity. Based on 15 RCTs including 2,351 participants, we identified several key findings. First, PN was associated with statistically significant, albeit modest, reductions in body weight and body fat compared with control interventions. Second, PN demonstrated no significant effects on BMI, waist circumference, lipid profile, or glycemic markers. Third, the included RCTs varied considerably in their approaches to PN implementation, particularly in the individual-level features considered, the ML algorithms applied, and the strategies developed and delivered for dietary personalization. Finally, nearly half of the trials (46.7%) either implemented specific dietary patterns or withheld dietary counseling from the comparison group. Taken

together, these findings indicate that the current evidence for PN remains mixed across outcomes and is shaped by substantial methodological variation across trials.

Challenges in Defining Appropriate Comparators for Personalized Nutrition Trials

A relevant consideration in interpreting the cardiometabolic findings is the variability in comparator conditions across the included RCTs. These control groups were not standardized but instead spanned a continuum of intensity, from minimal exposure such as generic dietary advice or educational materials to structured regimens involving prescriptive diets and guideline-based counseling. This variability is scientifically significant because the nature of the comparator fundamentally defines the research question being answered—whether PN demonstrates superiority over minimal guidance, or whether the personalization algorithm provides incremental benefit beyond an already established, evidence-based lifestyle intervention.

The first level consisted of minimal guidance, where control groups received passive, informational materials without active, individualized counseling. This approach included providing participants with educational pamphlets, a single leaflet outlining national food-based dietary guidelines [26, 29, 31, 34], or standard, non-personalized digital resources such as online materials and videos [11, 28, 32]. Examples of such national guidelines include the Dietary Guidelines for Chinese Residents [34], the Nutritional Guidelines for Filipinos [32], the food-based dietary guidelines published by the Netherlands Nutrition Centre [31], and the United States Dietary Guidelines for Americans (2020–2025) [11]. In these RCTs, the control condition represented a comparison against usual care or basic health education. The second level involved more structured, standard counseling where participants in the control group received a specific or generic diet plan. For example, one control group was prescribed a standard balanced diet as part of the established MOVE! weight management program for veterans [35], while others were instructed to follow a standardized low-fat diet and received group counseling sessions [25, 27, 33, 36]. The third and most intensive level involved high-intensity standard-care controls, where participants received structured and well-supported dietary and behavioral programs that closely approximated the level of intervention delivered in the PN arm [30]. In the PREVENTOMICS trial, for example, both the PN and control groups received easy-to-prepare meal boxes twice weekly, covering approximately 60% of total food intake. Notably, the authors reasoned that the control diet was likely healthier than the diet followed by participants in the PN group. The difference was that control participants received only non-personalized app-based messages [12]. Similarly, in the digital twin–

enabled PN trial in type 2 diabetes, control participants received comprehensive standard diabetes care, including regular physician visits and structured coaching at quarterly intervals [37]. The use of such active and well-supported control groups may have reduced the observable incremental effect of personalization in some trials, particularly for cardiometabolic outcomes, as the standard care provided was already a high-quality intervention [12, 25, 26, 32, 35].

Taken together, variability in the heterogeneity and intensity of control arms may have contributed to differences in observed effect sizes across trials. Clearer differentiation and standardization of comparator types are needed in future trials to enable valid interpretation of effect sizes, synthesis of evidence, and assessment of the true added value of PN beyond standard dietary interventions.

Divergent Strategies for Personalized Nutrition Delivery Across Trials

The implementation of PN across the 15 studies revealed notable differences in how personalization was applied, presenting distinct challenges and offering insights for future research. First, both the type and quantity of individual-level features employed to inform PN advice differed markedly across studies. Some trials, such as Food4Me and MyGeneMyDiet, tailored dietary recommendations using genetic and metabolic data to provide individualized nutrition guidance [26, 32]. However, other studies, like those using CGM-based personalization, provided feedback primarily based on participants' real-time metabolic responses to food [11, 25, 29, 30]. Additional approaches incorporated microbiome profiles [11, 25], anthropometrics [25, 26, 31, 32, 34, 37], physical activity [25, 26, 31-33, 37], medical history [11, 25], and dietary habits [11, 26, 31, 32, 34, 37] as guiding features. Notably, the number of each type of individual-level feature included in algorithmic models also varies considerably. For instance, as shown in **Supplementary S6**, Celis-Morales et al. and Jianheng Zheng et al. included only a single FTO gene [26, 34], whereas Mary S. McCarthy et al. incorporated 80 genes, and Mona A. Aldubayan et al. used 47 genes [12, 28]. This diversity reflects the absence of common criteria for selecting and integrating individual-level features, making it difficult to determine which personalization strategies yield meaningful effects or to synthesize findings across trials. In fact, one RCT suggests that the impact of the genotypic results may have been diluted by the large volume of other information provided [26]. This challenge is compounded by the inconsistency of reports across commercial nutrigenetic companies and the high cost of omics technologies, which further complicates the research landscape [35, 38]. Moreover, this lack of standardization

obscures mechanistic insights into how individual-level features interact with diet to shape cardiometabolic health. Future research should provide explicit justification for the selection of biological and behavioral features and undertake systematic comparisons of personalization strategies to identify approaches that generate the most robust evidence and hold the greatest potential for clinical and public health application.

Second, the delivery mode of PN interventions varied significantly across studies. Delivery modalities ranged from primarily digital, algorithm-driven applications that sent automated food-score nudges or decision tree prompts [11, 25, 34], to hybrid approaches that combined apps with scheduled counseling sessions or genotype-based feedback delivered by dietitians in person or via telehealth [32, 37], and even to resource-intensive models that supplied participants with up to 60% of their personalized food intake [12]. Interventions that layered frequent healthcare worker contact—such as weekly counseling, regular coaching calls, or in-person sessions—typically offered stronger implementation support and enhanced retention. However, these designs also introduced interpretive challenges, as differences in the amount of counseling contact between PN and control arms may confound the attribution of effects to personalization itself (e.g., mHealth pilot and digital-twin studies that combined intensive human coaching with sensor feedback) [30, 37]. This raises an interpretive issue: whether the observed effects reflect the personalization of individual-level features through algorithms, the intensity of delivery and support, or the interaction of both. Future PN trials will need to disentangle these elements by standardizing delivery intensity across arms or explicitly testing the added value of personalization beyond the effects of counseling.

Third, the duration of PN interventions varied substantially, ranging from a few weeks [12, 29, 33, 34] to year-long trials [32, 37]. Intervention length may influence whether PN effects are observed only in the short term or maintained over longer follow-up. For example, a 12-month trial found that PN produced greater weight loss at 3 and 6 months, but this advantage was not maintained at 12 months compared with standard care [37]. A similar pattern was observed in a 12-month study, in which early weight-loss progress was not sustained by the end of the trial [32]. Conversely, improvements in other clinical outcomes, such as HbA1c, were observed only in a 12-month trial, whereas a 24-week intervention yielded no such effects [30, 35]. This suggests that, for any short-term effects of PN on weight-related outcomes, longer follow-up is required to evaluate their sustained effects on cardiometabolic biomarkers and overall clinical relevance.

Collectively, PN implementations diverged in what individual-level features informed dietary personalization (e.g., genotype, biomarkers, behavioral habits, or multi-omic profiles), how

advice was delivered (e.g., app only, app plus human coaching, or counselor-led), and how long and intensively participants were supported. These differences may have contributed to variation in efficacy, adherence, and the comparability of trial outcomes.

Strengths and limitations of the review

This meta-analysis has several key strengths. All included studies were RCTs, providing methodological reliability. The use of predefined inclusion criteria, the GRADE approach, and subgroup and sensitivity analyses contributed to transparency in the evaluation process. Most trials were conducted in free-living, remote settings, enhancing the generalizability of findings to real-world conditions compared to traditional, tightly controlled clinical trials.

However, several limitations must be acknowledged when interpreting the results. First, although subgroup analyses were conducted across multiple study-level characteristics, substantial residual heterogeneity remained for body weight, BMI, and waist circumference, where residual I^2 remained high across most subgroup categories. For body fat, heterogeneity was reduced in selected subgroups but not consistently across all categories, suggesting that the examined variables explained only part of the between-study variability. This likely reflects differences in personalization methodologies across trials, which are difficult to avoid, as well as additional unmeasured or incompletely reported factors beyond the subgroup characteristics assessed that may have contributed to the heterogeneity observed in the main analyses. Second, restricting the review to peer-reviewed studies published in English may have excluded relevant non-English or unpublished evidence. Third, limited reporting on data collection methods and algorithms used for personalization reduced the ability to assess how PN strategies were designed, implemented, and evaluated. For example, one RCT noted that the details of its algorithm could not be disclosed because of a pending intellectual property rights application [12]. Finally, adherence to dietary interventions was inconsistently defined, measured, and reported across studies, with assessments ranging from self-reported dietary compliance to engagement with digital tools or attendance at counseling sessions, while several trials provided limited adherence data. As personalized nutrition depends not only on intervention design but also on the extent to which participants follow the prescribed

advice, this variability may contribute to differences in the magnitude of observed effects and their interpretation.

Conclusion

In summary, compared with controls, PN produced greater reductions in body weight and body fat but no additional benefits for BMI, waist circumference, lipid, or glycemic markers. Variability in the individual-level features guiding personalization, the modes of delivery, and the duration and intensity of participant support limits comparability across trials and hinders the identification of effective approaches. Currently, PN faces obstacles such as limited reproducibility, methodological inconsistencies in study design, and the complexities of analyzing and interpreting high-dimensional data. While advances in sequencing technologies are expected to lower costs, the absence of a clear, standardized definition of PN and a robust framework for its application remain key barriers to its effective integration into clinical practice. To improve methodological consistency and interpretability, future research should establish consensus definitions of PN and develop classification frameworks that specify individual-level features, personalization mechanisms (e.g., algorithmic vs. clinician-guided), intervention intensity and duration, the use of well-matched comparative control groups, and transparent reporting of data collection methods and personalization algorithms. In light of current evidence, routine clinical implementation of PN cannot be broadly recommended until further studies establish which strategies are both effective and scalable for improving cardiometabolic health.

CRediT authorship contribution statement

1. TQD: Conceptualization, Methodology, Data curation, Investigation, Visualization, Writing – original draft, Writing – review & editing.
2. NDK: Conceptualization, Investigation, Writing – review & editing.
3. DDK: Software, Validation, Investigation, Writing – review & editing.
4. NTTH: Investigation, Visualization, Writing – review & editing.
5. NVK: Software, Validation, Data curation, Writing – review & editing.
6. TTT: Software, Validation, Investigation, Writing – review & editing.
7. All authors: All authors reviewed the results, critically revised the manuscript, and approved the final version for submission.

Funding

The authors reported no funding received for this study.

Conflict of interest

The authors report no conflicts of interest.

Acknowledgements

Not applicable.

Data Availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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Figure Legend

Figure 1. Study characteristics. Abbreviations: CGM, continuous glucose monitoring; ns, non-significant.

Figure 2. Forest plots displaying the effect of personalized nutrition in adults on (A) body weight (kg), (B) body mass index (kg/m²), (C) body fat (%), and (D) waist circumference (cm). Abbreviations: CI, confidence interval; MD, mean difference; SD, standard deviation; Total, total number of participants who completed the study.

Figure 3. Forest plots displaying the effect of personalized diet in adults on (A) fasting glucose (mg/dL), (B) Insulin (μ IU/mL), and (C) HbA1c (%). Abbreviations: CI, confidence interval; MD, mean difference; SD, standard deviation; Total, total number of participants who completed the study.

Figure 4. Forest plots displaying the effect of personalized diet in adults on (A) total cholesterol (mg/dL), (B) low-density lipoprotein cholesterol (LDL-C, mg/dL), (C) high-density lipoprotein cholesterol (HDL-C, mg/dL), and (D) triglycerides (mg/dL). Abbreviations: CI, confidence interval; MD, mean difference; SD, standard deviation; Total, total number of participants who completed the study.

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